STAFF REVIEW · H446 087 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A condition maintained while processing data.

2. A single scan can find it in O(n).

3. An int? that can be an integer or null.

4. mid2=6, then mid3=8; found index3.

I Do — source worked steps

1. Model response — The invariant tracks largest seen, not last examined.

We Do — guided CHECK

1. Model response
Starting best at0 would be incorrect for all-negative arrays.

You Do — source mark schemes

1. Initial−5 [2], becomes−2 [2], remains−2 [1], returns−2 [1].

2. Empty return null [2], initial best=a[0] [1], iterate array [1], update if x>best [2], return best [1].

3. Scan O(n) [2], sorting typically O(n log n) [2], test empty and all-negative cases [2].

Exit Ticket — source answer

1. False. It fails on arrays containing only negative values.